

Title: Image-Based Soil Type Classification Using Multi-Feature Fusion and Classical Machine Learning Algorithms.

Abstract

Soil classification is a significant area of study in agricultural production, land use planning, and environmental monitoring. This study proposes an automatic classification method utilizing image processing and machine learning, derived from six soil image categories—Black Soil, Cinder Soil, Laterite Soil, Peat Soil, Loam Soil, and Yellow Soil—sourced from the Kaggle dataset. We made a composite feature vector by taking color features, Haralick texture features, shape descriptors, Local Binary Pattern (LBP) features, surface features, and statistical features from each image. The classification process uses three supervised machine learning models: Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and Decision Tree (DT). We used metrics like accuracy, precision, recall, F1-score, and classification success index (CSI) to judge how well the models worked. The experimental results show that the Decision Tree model is the most stable and consistent compared to other models, while KNN achieved the highest accuracy in a specific feature setting. This means that the proposed multi-feature-based framework with a tree-based classifier works better. The findings of this study indicate that the integration of diverse visual and textural features in image-based soil identification can markedly enhance classification accuracy. This method will be useful in the future for making automated agricultural decision support systems, smart farming, and land resource management technologies.

1. Introduction

Soil is a vital natural resource for agriculture, infrastructure development, and environmental management, significantly influencing crop yield, water retention, nutrient availability, and soil stability. Bangladesh's varied geomorphological and geological settings result in notable disparities in the physical and mineral characteristics of soils across different regions. The soils of the Barendra region, haor areas, coastal lands, and hilly regions are very different from each other in terms of color, particle size, structure, organic matter content, mineral content, and surface texture [1]. Farmers and the general public find it challenging to accurately figure out these properties just by looking at them. Choosing the wrong type of soil can lead to the wrong crops, using too much fertilizer, wasting water, and long-term damage to the land [2], [3], [4]. So, it is essential to quickly and accurately identify the type of soil to make sure that agriculture and land management are safe and sustainable.

In developing nations such as Bangladesh, conventional soil testing procedures that rely on chemical and geological investigations are frequently prohibitively expensive, time-consuming, and difficult to access in remote regions. In this context, it is essential to develop a rapid, cost-effective, and automated method for classifying soil using photographs. Recently, there has been a rapid increase in the application of computer vision and machine learning to extract color, texture, and structural elements from soil photographs and categorize them.

In recent time, Shuaeb *et al.* [5] offered new method to detect soil wetness. The study used four machine learning and CNN models. Among these models CNN demonstrates the highest result. On the other hand, image analysis-based methods use soil color and texture as key indicators to determine the dryness status. Soil color and appearance are indicative of many physical and chemical properties, although they may vary due to environmental changes, location, or natural processes. Recent studies have attempted to classify soil moisture using image analysis, showing promise as a fast, cost-effective, and non-invasive alternative to traditional methods [6], [7]. In addition, Acharjee *et al.* [8] offered a machine learning model to predict crop yield and soil moisture. On the other hand, RGB image processing has been used for iron and carbon prediction from soil images. The moisture content of the soil was classified using the CIE L*a*b* mode, and the depth of soil was estimated by L*a*b* mode [9]. The scheme of [10] offered soil water content accuracy in sandy Loamy soil. Asensio *et al.* [11] proposed model to forecast eighteen soil characteristics, including carbon, nitrogen, iron, sand, and clay, using RGB image processing. Another soil texture image classification as suggested by [12]. RGB histogram analysis was used to classify the soil surface image in this study. Jiao *et al.* [13] introduced a chip less soil moisture sensor device called SoilTAG, which uses battery-free inactive tags to determine soil moisture levels. Moreover, Liu *et al.* [14] offered an analysis approach based on support vector machines (SVM) to identify the urban soil. Machbah *et al.* [15] developed a novel feature-based machine learning model to classify the soil type. A study published in Scientific Reports in 2025 [16] proposed a hybrid soil classification framework combining handcrafted textural features and deep representations, which shows promising results for precision agriculture. However, most of these studies rely on images captured in controlled environments or remote sensing data, so the robustness of the models to changing light, humidity, and background conditions in real fields has not yet been adequately validated. In addition, the lack of open and standardized image datasets for specific classes such as Black, Cinder, Laterite, Peat, Loam, and Yellow soil makes it difficult to compare the results of different studies. At the same time, there is a potential for low-resource, interpretable decision trees or feature-based models to replace computationally expensive deep learning methods, but systematic comparative research in this area remains limited. In this context, the present study proposes an automatic classification framework comprising color, Haralick texture, shape, local binary pattern, surface and statistical features using RGB images of six soil classes—black, cinder, laterite, peat, loam and yellow—collected from Kaggle. The classification performance was verified through comparative analysis of Support Vector Machine, K-Nearest Neighbor and Decision Tree models, where Decision Tree provided the best results. This study can be considered as an important step towards the development of a field-usable, low-cost and interpretable soil classification system. The main contribution of the paper is as follows:

- i. A multi-feature classification framework combining color, Haralick texture, shape, LBP, surface and statistical features is proposed using Kaggle-based six-class soil images.
- ii. A comparative analysis of SVM, KNN and decision tree classifiers is performed using accuracy, precision, recall, F1-score and CSI metrics, where the decision tree gives the best results.
- iii. The potential of an interpretable soil classification system usable in low-resource environments is demonstrated.

The remainder of the paper is organized as follows: Section 2 details the methodology, Section 3 discusses the experimental results, and Section 4 presents the conclusion.

2. Methodology

This study proposes a step-by-step framework for soil image-based automatic classification, which includes the complete process of image acquisition, pre-processing, feature extraction, feature vector construction, machine learning-based classification, and performance evaluation. The overall workflow of the proposed system is presented in Figure 1, where the interrelationships and information flow of each step are clearly displayed.

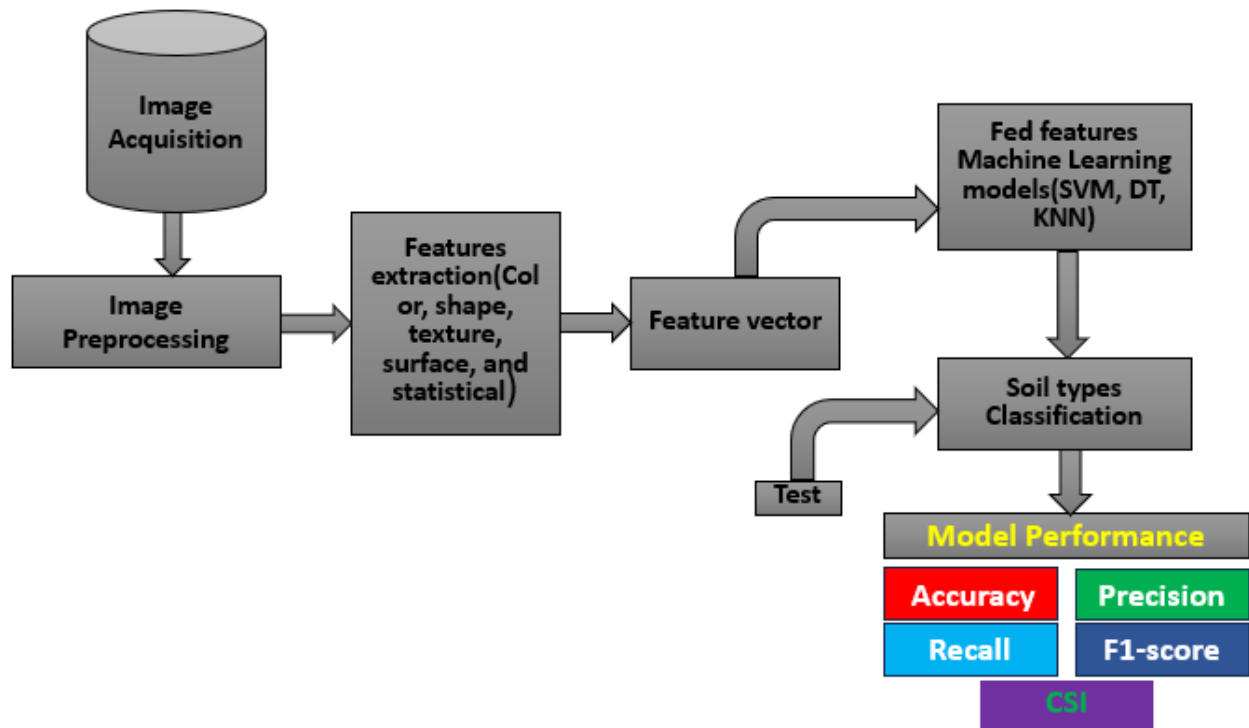


Figure 1. Represents the proposed system.

2.1 Image Acquisition: In the first step, images of six soil types—Black, Cinder, Laterite, Peat, Loam, and Yellow—are collected from the Kaggle dataset. These images are stored in RGB format and contain a variety of lighting conditions, surface textures, and colors, which are important for real-world simulations.

2.2 Image Preprocessing: Soil images captured in laboratory environments may contain various types of noise, which can negatively affect the performance of feature extraction and classification

models in the subsequent steps. Furthermore, the relatively large size of images captured by high-resolution cameras such as iPhones creates additional computational complexity when used directly. To address these issues, a structured image pre-processing step is followed. The implemented steps are shown in Figure 2(a)–(e).

2.2.1 Resizing: In the first step, each image is scaled to 250×250 pixels to convert it to a uniform size. This allows for unification of the input data and facilitates data processing in the machine learning model.

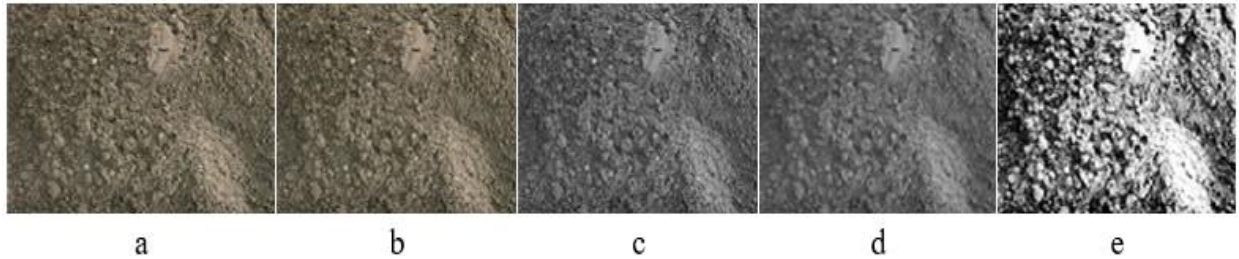


Figure 2(a)–(e): Sequential illustration of the image pre-processing pipeline, showing the original RGB image (a), resized image at 250×250 pixels (b), grayscale conversion (c), noise reduction using median filtering (d), and contrast enhancement through histogram equalization (e).

2.2.2 Color Conversion: In the next step, the RGB color images are converted to grayscale. This conversion reduces unnecessary color-based variations and reflects texture and structural features relatively clearly, which plays a helpful role in feature extraction.

2.2.3 Noise Reduction: A median filter is applied to improve image quality, which has been reported in previous studies to be able to effectively reduce speckle noise and impulse-type distortion. This step reduces noise while preserving important edge and textural information[16].

2.2.4 Contrast Enhancement: The final step uses histogram equalization to increase the pixel quality of the image and to more clearly present fine details[17]. This technique improves the overall contrast of the image and makes the textural pattern more visible, which serves as an important input for classification. The above preprocessing steps transform the raw soil images into high-quality, model-useful inputs, which helps ensure improved performance in the subsequent feature extraction and classification steps.

2.3 Feature Extraction: In this step, a composite feature representation is constructed by measuring various visual features from pre-processed soil images, which is used as input for the next classification step. Since different types of soils exhibit significant differences in color, particle structure, surface roughness, and textural properties, an attempt is made to increase the classification performance by using multiple complementary feature groups together[18]. A total of six classes of features are extracted in this study, which are explained in detail below.

2.3.1 Color Features: Color-based features play an important role in determining soil-type differences, because the color distribution differs in different soils according to the amount of mineral content and organic matter. In this study, the mean, standard deviation, and histogram-based indices are calculated from RGB and grayscale channels, which represent the overall color characteristics of each image[19].

2.3.2 Shape Features: Shape features indicate the geometric arrangement of soil particles or surfaces and reflect surface asperity, edge density, and structural differences. This study used contour-based measurements, area, perimeter, aspect ratio, and compactness to characterize the geometric properties of the surface[20].

2.3.3 Texture Features (Haralick / GLCM-based): Textural features are particularly useful in identifying the fine structure of the soil. Haralick's proposed features—such as contrast, correlation, energy, and homogeneity—were calculated using the Grey Level Co-occurrence Matrix (GLCM), which analyses the correlation between neighboring pixels to indicate the rough or smooth nature of the soil surface[15], [21].

2.3.4 Local Binary Pattern (LBP) Features: LBP is a local texture description method, where each pixel is compared to its neighboring pixels to create a binary pattern. Histogram-based indices were extracted from these patterns to identify small-scale texture changes on the soil surface, which complements Haralick features[22].

2.3.5 Surface Features: Surface features indicate the roughness, pitting, or high-low structure, and overall structural nature of the soil surface. In this study, the surface texture diversity is expressed numerically by measuring gradient-based indices, edge density and local variation[23].

2.3.6 Statistical Features: Statistical features represent the overall distribution of pixel values in the image. Here, the tonal characteristics of each image are summarized by calculating indices such as mean, variance, standard deviation, skewness and kurtosis. A composite statistical feature vector is created for each soil image by combining the above six types of features, which is used as input to machine learning models. This multidimensional feature representation helps to identify subtle differences between different soil classes and plays an important role in increasing the overall accuracy of classification.

2.4 Feature Vector Construction: In the feature extraction step, six types of features—color, shape, texture (Haralick/GLCM), local binary pattern (LBP), surface, and statistical—are expressed numerically from each soil image. However, instead of using these features separately, it is necessary to form a unified and high-dimensional representation for each image, which can be effectively used in the subsequent classification step. For this purpose, a single feature vector is created by connecting all the feature groups, which contains the multidimensional features of the corresponding soil image together. First, each feature set is transformed to a certain number of dimensions, and all the features are brought to a uniform numerical range by applying scaling or normalization techniques so that no single feature group has an undue influence. This step is particularly important, because the different ranges of values of different features can cause bias in the training process of the machine learning model. Then, a unified feature vector is formed by concatenating each feature group sequentially. This vector is of fixed length for each image so that all samples in the dataset maintain the same structure and are suitable for use as input to the model. The resulting feature vector carries information such as color and textural information on the one hand and includes morphological and surface-based features on the other, which increases the ability to detect subtle differences between different soil types. In this study, dimensional consistency has been maintained during feature vector construction, and, if necessary, feature selection or reduction techniques have been applied to eliminate redundant or less important features so that the complexity of the model is controlled and the generalization ability is improved. The final feature vector is provided as input to the Support Vector Machine (SVM), K-Nearest Neighbor (KNN) and Decision Tree (DT) models in the next step, through which soil image classification is completed. This integrated feature representation serves as an important component of the proposed framework, as it helps to increase the accuracy and stability of classification by combining different types of visual features.

2.5 Classification Models

The numerical representation of each soil image prepared in the feature vector construction step is used as input in the subsequent classification step. In this study, three widely used supervised machine learning algorithms—Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and Decision Tree (DT)—are selected to comparatively analyze the performance of different learning techniques on the feature-based soil classification problem. First, the entire dataset is divided into

training and testing subsets to properly evaluate the generalization ability of the models. In the training step, each model learns using the feature vectors and the corresponding class labels and constructs a decision tree that classifies according to soil type. On the other hand, in the testing step, the real-world performance of the models is verified by performing classification on previously unseen data.

In the case of Support Vector Machine (SVM), classification is accomplished by determining the best separating hyperplane between classes in a high-dimensional feature space, which is effective in detecting complex and non-linear patterns. In the K-Nearest Neighbor (KNN) method, classes are determined based on the K number of training samples closest to each test sample, which is helpful in analyzing local feature similarity. On the other hand, the Decision Tree (DT) algorithm divides the feature space through a series of decision rules to create an interpretable classification structure, where the most informative feature is selected at each node[4], [24], [25]. A similar testing environment has been maintained in these three models by applying the same feature vector set so that the comparative analysis of the results is fair and reliable. Through this, it has been possible to determine which algorithm performs best with the proposed multi-feature representation.

2.6 Model Evaluation

The performance of the model is evaluated using four standard metrics. This step makes sure that the model can correctly classify the levels of dryness in Loamy soil using image-based analysis so that the soil condition rating can go more smoothly[26], [27], [28], [29]. The four standard metrics are discussed respectively.

Accuracy: Overall correctness of predictions. Mathematically,

$$\text{Accuracy} = (TP + TN)/(TP + TN + FP + FN) \quad \text{Eq. (1).}$$

where TP, TN, FP and FN represent the true positive, true negative, false positive, and false negative respectively.

Precision: Correct positive predictions. Mathematically,

$$\text{Precision} = TP/(TP + FP) \quad \text{Eq. (2).}$$

Recall: Ability to detect all positives. Mathematically,

$$\text{Recall} = TP/(TP + FN) \quad \text{Eq. (3).}$$

F1-score: Balance of precision & recall. Mathematically,

$$\text{F - Score} = (2 * (\text{recall} * \text{precision})) / (\text{recall} + \text{precision}) \quad \text{Eq. (4).}$$

Classification Success Index (CSI): The Classification Success Index evaluates the proportion of correctly classified positive instances relative to the total number of samples that were either predicted or actually positive. This metric is particularly useful in multi-class and imbalanced classification scenarios:

$$\text{CSI} = TP/(TP + FP + FN) \quad \text{Eq. (5).}$$

3. Result Analysis

In this section, the steps of dataset preparation used and the performance of the proposed machine learning models are discussed in detail. Specifically, the accuracy, precision, recall, F1-score and classification success index (CSI) obtained using different feature sets for soil types classification are analyzed comparatively. In addition, the overall performance of different classifiers is evaluated through the 10-fold cross-validation method, and the stability and generalization ability

of the models are highlighted using the precision–recall curve. In addition, which feature sets have the most impact on classification is analyzed, and the performance and potential contribution of the proposed framework are clearly presented by comparing the results with recent research.

3.1 Dataset Preparation and Representation

The soils image dataset used in this study was collected from the Kaggle platform, as previously discussed in Section 2.1 (Image Acquisition). Initially, 100 original RGB images were included for each soil class, resulting in a total of 600 initial images for six classes—Black, Cinder, Laterite, Peat, Loam, and yellow soil.



Figure 3: Illustration of the representative samples of the different soil types considered in this study.

Table 1 demonstrate the augmentation techniques applied to the soil type dataset

SN	Soil Types	Original Image	Rotation (angles 90,120, 270 to180	Scaling (up-.8, ,1.2 down)	Reflection (Horizontal-Vertical)	(Total Original + Augmentation Image)
1	Black Soil	100	4×100	4×100	2×100	1100
2	Cinder Soil,	100	4×100	4×100	2×100	1100
3	Laterite Soil,	100	4×100	4×100	2×100	1100
4	Peat Soil	100	4×100	4×100	2×100	1100
5	Loam Soil	100	4×100	4×100	2×100	1100
6	Yellow Soil	100	4×100	4×100	2×100	1100
The dataset contains the total number of soil type images						6600

To increase the diversity of the dataset and improve the generalization ability of the models, a data augmentation technique was applied. In this case, multiple geometric transformations were performed on each original image. First, the images were rotated at different angles, specifically at 90°, 120°, 180°, and 270°, thereby creating an additional 4×100 images for each class. Second, the image size was changed by a factor of 0.8, 0.9, 1.0, 1.2, and 1.3 through scaling operations, which added 4×100 additional augmented images per class. Third, an additional 2×100 images per class were created by applying horizontal and vertical reflection. By applying all these augmentation techniques together, a total of 1100 images were produced for each soil class. As a

result, the size of the complete dataset for the six classes was 6600 soil images, which is presented in Table 1. This augmented dataset provided a strong basis for reliably evaluating the performance of the machine learning models in the subsequent training and testing phases.

3.2 Explanation of Different Experimental Settings

In Table 2, the classification performance of Support Vector Machine (SVM), Decision Tree (DT) and K-Nearest Neighbor (KNN) models on different feature sets is compared. Here, the results of each test are presented using accuracy, precision, recall, F1-score and the classification success index (CSI). It can be seen from the table that the performance of most models is relatively low when using a single feature (such as only Shape, Surface or Statistical). In contrast, when using multiple features together (color texture, Color & Shape or all features), the classification accuracy is significantly increased. This indicates that the combination of different features plays a very important role in soil image classification.

Table 2. Performance scores for different feature sets on the augmented image dataset

Ex.N	Feature Set	Model	Accuracy	Precision	Recall	F1-Score	CSI
1	only Color	SVM	0.816	0.811	0.816	0.805	0.627
		DT	0.93	0.932	0.93	0.93	0.862
		k-NN	0.922	0.924	0.922	0.922	0.846
2	Texture_GLCM	SVM	0.503	0.519	0.502	0.489	0.022
		DT	0.807	0.813	0.807	0.808	0.62
		k-NN	0.813	0.816	0.813	0.813	0.629
3	Texture_LBP	SVM	0.654	0.659	0.654	0.649	0.313
		DT	0.816	0.815	0.816	0.815	0.631
		k-NN	0.83	0.831	0.83	0.829	0.661
4	Texture_GLCMLBP	SVM	0.701	0.701	0.701	0.695	0.402
		DT	0.846	0.849	0.846	0.846	0.695
		k-NN	0.849	0.852	0.849	0.849	0.701
5	Color Texture	SVM	0.869	0.871	0.869	0.867	0.74
		DT	0.908	0.908	0.908	0.908	0.816
		k-NN	0.902	0.904	0.902	0.902	0.806
6	Color & Shape	SVM	0.86	0.86	0.861	0.859	0.721
		DT	0.933	0.934	0.933	0.933	0.867
		k-NN	0.939	0.939	0.939	0.938	0.878
7	Shape Texture	SVM	0.768	0.774	0.768	0.766	0.542
		DT	0.849	0.854	0.849	0.85	0.703
		k-NN	0.849	0.851	0.85	0.848	0.701
8	Shape Statistical	SVM	0.799	0.802	0.799	0.797	0.601
		DT	0.888	0.894	0.889	0.888	0.782
		k-NN	0.877	0.878	0.877	0.876	0.755
9	Surface	SVM	0.589	0.607	0.589	0.573	0.196
		DT	0.807	0.81	0.807	0.807	0.617
		k-NN	0.788	0.797	0.788	0.786	0.584
10	Shape	SVM	0.547	0.589	0.547	0.543	0.136
		DT	0.785	0.788	0.785	0.783	0.573
		k-NN	0.774	0.776	0.774	0.774	0.55
11	Statistical	SVM	0.67	0.675	0.67	0.657	0.346
		DT	0.894	0.895	0.894	0.894	0.789
		k-NN	0.88	0.884	0.88	0.879	0.764
12	All_Features	SVM	0.913	0.915	0.914	0.913	0.828
		DT	0.93	0.931	0.93	0.93	0.861
		k-NN	0.927	0.929	0.927	0.928	0.856

Only Color (Setting- 1): All three models performed well using the color feature set, where DT achieved the highest accuracy (0.93), while KNN showed very similar performance. This indicates that soil color is a highly effective feature for soil classification.

Texture-GLCM and Texture-LBP (Settings 2 and 3): Using only GLCM or LBP features noticeably reduced the performance of SVM. However, DT and k-NN performed comparatively better on these feature sets. In the Texture_LBP setting, k-NN achieved the highest accuracy (0.83). These findings suggest that texture-based features alone provide limited classification performance, although they remain moderately effective for soil classification.

Shape, Surface and Statistical (Setting 8–11): The classification performance decreased when Shape, Surface, and Statistical features were used individually. Among the classifiers, SVM showed comparatively weaker performance, while DT and k-NN achieved moderate results. Among these feature sets, Statistical features produced relatively better performance, where DT achieved the highest accuracy of 0.894. These findings suggest that shape, surface, or statistical features alone are not sufficiently effective for accurate soil classification and are more useful when combined with other discriminative features.

Combined Feature Experiments: Color & Shape (Setting-6): This setting achieved the best overall performance in the experiment. k-NN obtained the highest accuracy (0.939) with a CSI value of 0.878, while DT also demonstrated excellent performance with an accuracy of 0.933. These results indicate that combining color and shape information provides highly discriminative features for soil classification.

All Features (Setting-12): When all extracted features were combined, the classifiers achieved highly stable and competitive performance. DT achieved an accuracy of 0.93, while k-NN achieved 0.927 accuracy with strong F1-Score and CSI values. These results indicate that integrating all available features enhances the overall classification capability of the system.

Most Influential Feature Sets: From the table analysis, color features showed strong classification capability when used alone. Texture features (GLCM and LBP) provided moderate performance individually but improved the results when combined with other features. Figure 4 represents the most influential feature set. Therefore, Color & Shape can be considered the most influential feature combination for soil classification, achieving the highest overall performance among all feature sets.

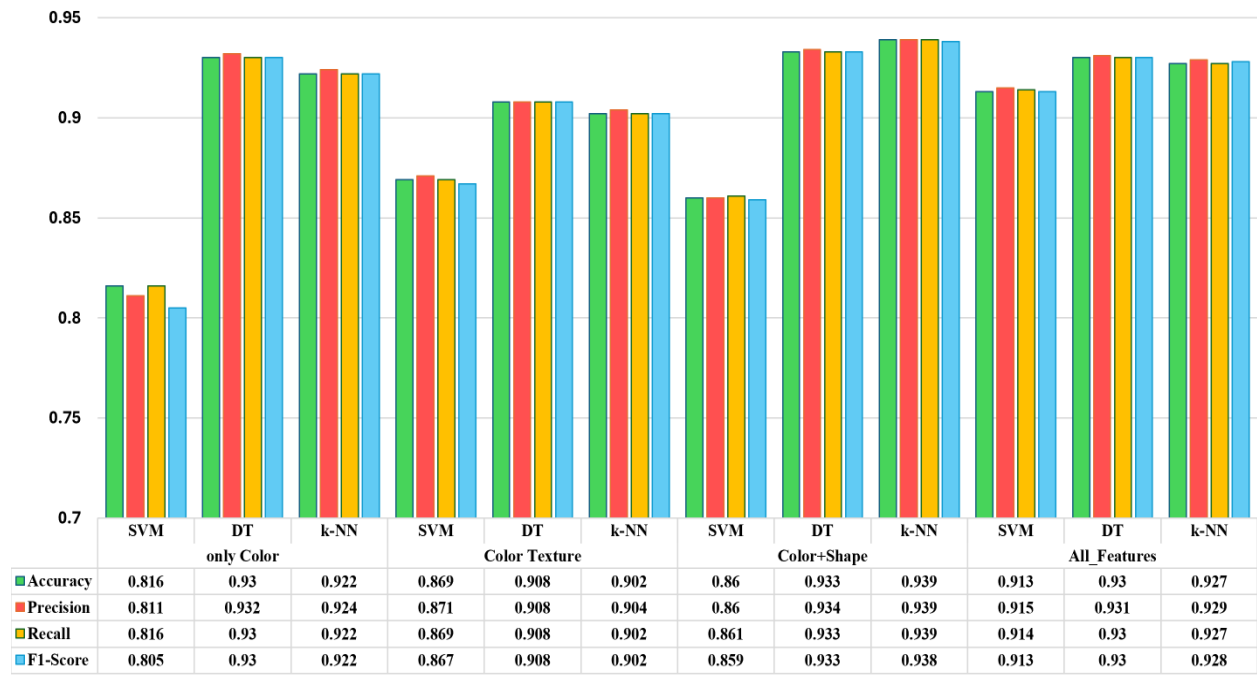


Figure 4. Most Influential Feature Groups.

Determining the Best Performing Model: All settings are combined: Decision Tree (DT) gives the highest or close performance in almost every test. KNN is the second-best performer. SVM lags behind in most cases, especially in the texture-only setting.

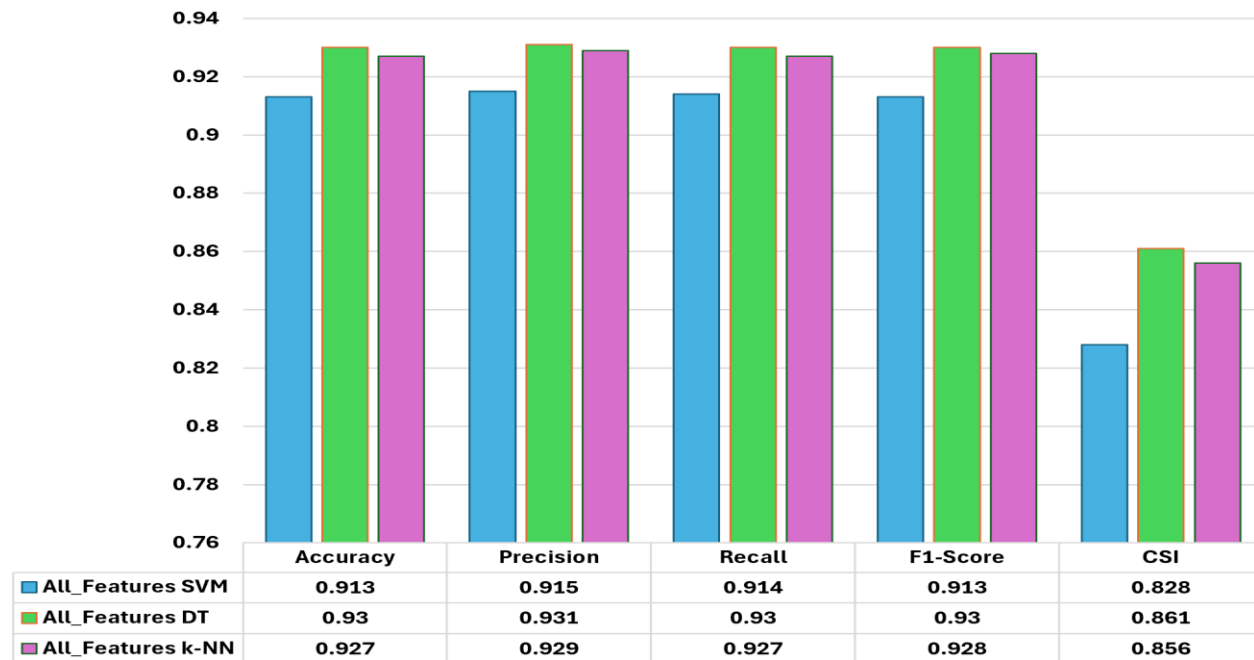


Figure 5. Depicts the Best model selection.

Therefore, the decision tree is selected as the best and most stable model in this study. Figure 5 indicates that the rule-based tree structure is capable of effective classification. Across all twelve experimental settings, decision tree consistently delivered the highest or near-highest performance among the three evaluated models. KNN ranked as the second-best performer, whereas SVM generally lagged behind, particularly in texture-only and geometry-only configurations. Based on this consistent superiority, decision tree is selected as the most reliable and stable classifier in the present study. The results further suggest that its rule-based hierarchical structure is well-suited for handling multi-feature soil representations and provides interpretable decision pathways, which is highly desirable for real-world agricultural and field-level applications.

Model Robustness Analysis:

In this study, as shown in figure 6, the Decision Tree (DT) model using All Features consistently performed well at different training-testing split ratios. The highest accuracy of 0.933 was achieved at 90:10 split ratio. In addition, the accuracy at 80:20 and 70:30 ratios were 0.931 and 0.928, respectively, indicating high stability of the model. The accuracy gradually decreased as the amount of training data decreased. While the accuracy was 0.914 at 60:40 ratio, the performance started to decrease further at 50:50 ratio and dropped to 0.688 at 20:80 ratio. However, this decrease was gradual, meaning the model did not completely fail and maintained reasonable performance. The results show that the DT model, although sensitive to the amount of training data, exhibited good generalization capability due to the use of All Features. The model provided the most reliable results, especially at 70:30, 80:20, and 90:10 split ratios. Among these, 80:20 and 70:30 ratios can be considered more suitable in terms of accuracy and evaluation balance. Figure 7 8 shows that the decision tree model was able to correctly classify most of the samples among the six soil classes. The high quality of the diagonal elements indicates that the prediction accuracy of the model is satisfactory. The highest correct classification was observed in the Loam Soil class, where 59 out of 60 samples were correctly predicted. Similarly, the model also performed well in the Black Soil, Peat Soil and Yellow Soil classes.

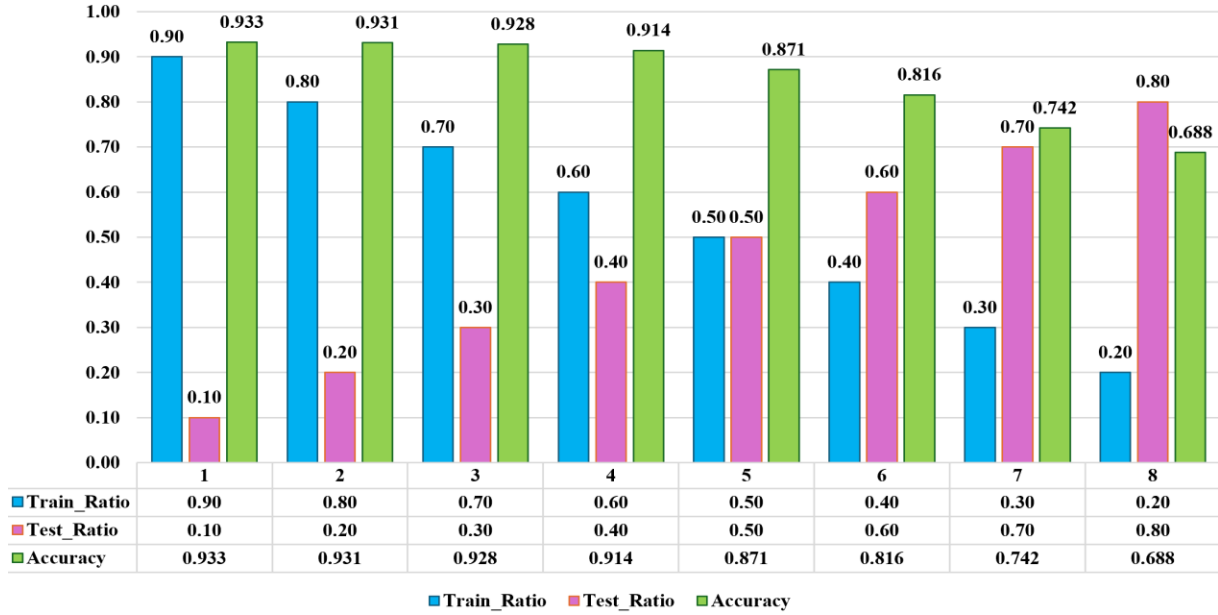


Figure 6. Robustness analysis of the proposed model under varying training-testing splits.

However, some misclassification was observed, especially in the Cinder Soil, Laterite Soil, Peat Soil and Yellow Soil classes. This may be due to the similarity in color, texture or surface pattern of these soil classes. Laterite Soil was predicted as Peat Soil and Loam Soil in some cases, which indicates that the extracted features of these classes were somewhat overlapping.

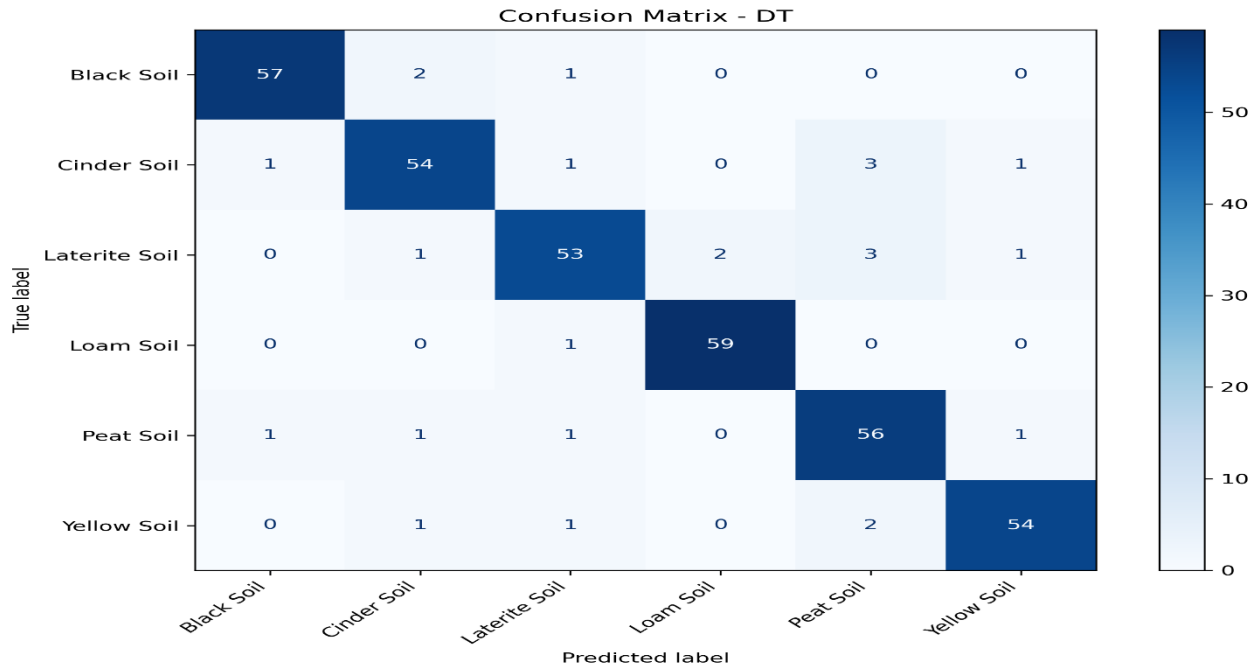


Figure 7. Represents the best model confusion matrix.

Overall, the confusion matrix demonstrates that the DT model provided robust and effective performance in the soil image classification task. Since most of the predictions are concentrated in the diagonal region, the class discrimination ability of the model can be considered good.

ROC Curve and AUC Analysis

In this study, the figure 8 shows the multiclass ROC curve of the Decision Tree (DT) model as the Best Model. The ROC curve is used to evaluate the classification capability of the model, where the relationship between the False Positive Rate (FPR) and the True Positive Rate (TPR) is shown. The closer the ROC curve is to the upper-left corner, the better the class separation capability of the model is considered. Figure 9, it can be seen that the ROC curves of the DT model are located much above the random classifier line. The random classifier line is shown by the diagonal dashed line, whose performance is usually close to $AUC = 0.50$. But here the AUC value of each class of the DT model is more than 0.93, which indicates that the model is able to distinguish soil classes very effectively. The Micro-average AUC = 0.9581 and Macro-average AUC = 0.9581 of the Best DT models were found. The similarity of micro and macro AUCs indicates that the overall performance and class-wise balanced performance of the model are almost the same. That is, the model is not biased towards any particular class; rather, it provides relatively stable performance for all classes.

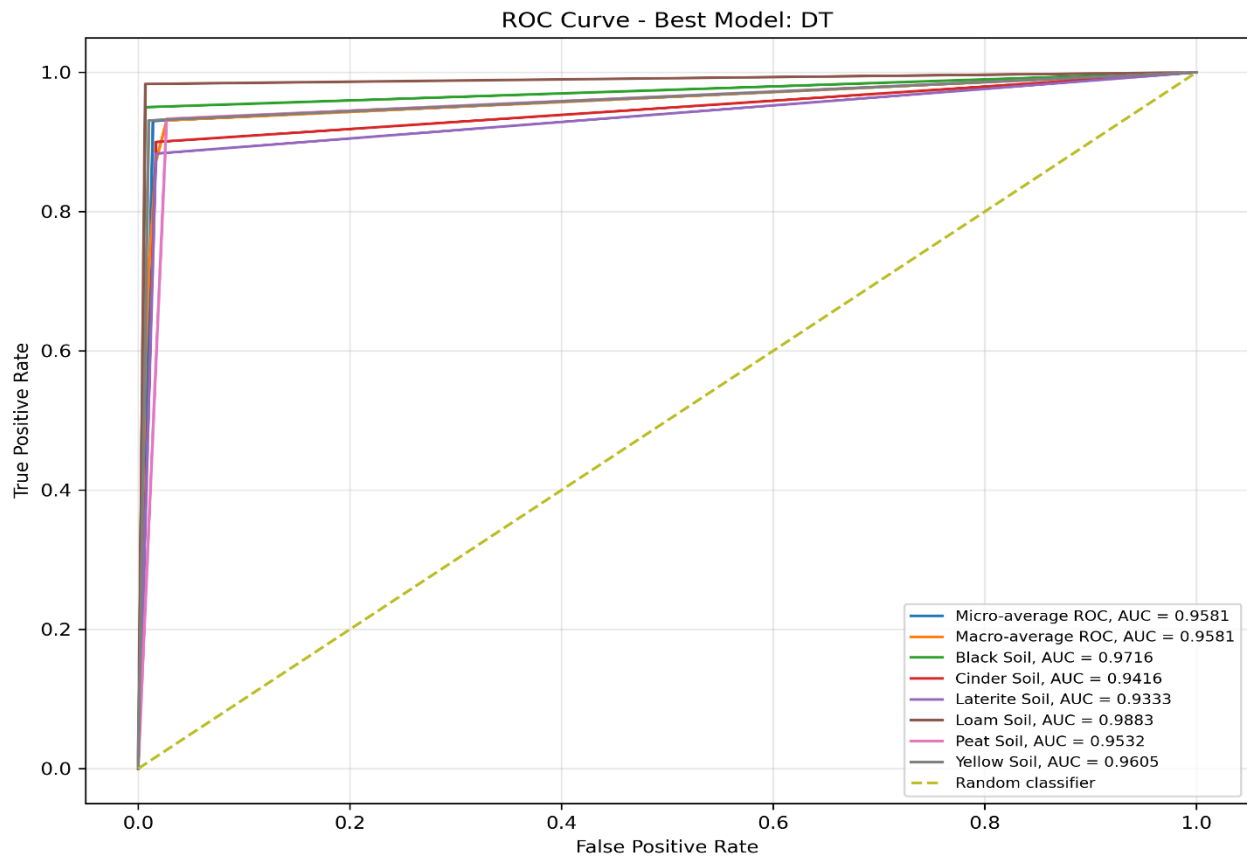


Figure 8. Demonstrates the ROC-AUC analysis of DT best model.

According to the class-wise AUC result, the Loam Soil class achieved the highest AUC, which is 0.9883. This indicates that Loam Soil is very well separated from other soil classes. Then, Black Soil and Yellow Soil achieved AUCs of 0.9716 and 0.9605, respectively, indicating strong classification performance. The AUC of Peat Soil class was 0.9532, the AUC of Cinder Soil class was 0.9416, and the AUC of Laterite Soil class was 0.9333. Although the AUC of Laterite Soil is relatively low, it is still more than 0.93, indicating that the discrimination capability of the model is satisfactory. Overall, ROC-AUC analysis demonstrates that the Decision Tree model exhibits highly effective and robust performance in the soil image classification task. By using All Features, the model is able to learn meaningful patterns from the color, texture, shape, statistical, and surface

characteristics of the soil classes. As a result, the model is able to maintain a high true positive rate while keeping false positives low.

4. Conclusion

In this study, a multi-feature-based machine learning framework was proposed for the automatic classification of six soil types—Black, Cinder, Laterite, Peat, Loam, and Yellow—using soil image datasets collected from Kaggle. A composite feature vector consisting of color, texture (GLCM and LBP), shape, surface, and statistical features was used to evaluate the performance of Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and Decision Tree (DT) classifiers. The experimental results demonstrated that combining multiple complementary features significantly improved classification performance compared with using individual feature sets. Among the investigated feature combinations, the Color & Shape feature set achieved the highest classification accuracy, indicating its strong discriminative capability for soil type identification. Although KNN achieved the highest overall accuracy, the Decision Tree model exhibited the most consistent and stable performance across different experimental settings. Furthermore, the interpretable nature of the Decision Tree classifier makes it particularly suitable for practical agricultural decision-support and soil assessment applications. Overall, the proposed framework demonstrates that effective feature fusion combined with classical machine learning algorithms can provide a reliable and cost-effective solution for image-based soil classification. Future work may focus on incorporating larger field-level datasets, evaluating deep learning approaches, and deploying the framework on real-time mobile or IoT-based platforms.

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Data Availability: <https://www.kaggle.com/datasets/prasanshasatpathy/soil-types>

Authors contribution statement

Md. Sakibur Rahaman: Conceptualization, Data curation, Formal analysis, Methodology, Resources, Software, Writing – original draft, and editing.

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